

Explain why you line up the decimal points to add $3.6 + 2.45$.

Sum: _____

Explanation: _____

Which describes a place-value model for $1.3 - 0.45$?

- A.** Add 4 tenths and 5 hundredths to 1 one and 3 tenths.
- B.** Start with 1 one and 3 tenths, trade 1 tenth for 10 hundredths, then take away 4 tenths and 5 hundredths.
- C.** Take away 1 one and 3 tenths from 4 tenths and 5 hundredths.
- D.** Take away 4 ones and 5 tenths from 1 one and 3 tenths.

True or false?

To find $0.7 - 0.25$, you can rename 0.7 as 70 hundredths and subtract 25 hundredths to get 45 hundredths.

- A.** True
- B.** False

Raj added $4.2 + 1.35$ and got 1.77.

Find the correct sum and explain his mistake.

Correct sum: _____

Explanation: _____

SET 62 • 5.NBT.B.7

BACKWARD 1

Add tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{4}{10} + \frac{25}{100}$$

Sum: _____

True or false?

$$\frac{3}{10} = \frac{30}{100}$$

A. True

B. False

$$\frac{8}{10} + \frac{7}{100} = ?$$

A. $\frac{87}{100}$

B. $\frac{15}{100}$

C. $\frac{15}{110}$

D. $\frac{87}{10}$

Write $\frac{6}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{2}{10} + \frac{9}{100} = \frac{11}{100}$$

A. True

B. False

SET 62 • 5.NBT.B.7

BACKWARD 2

Rename a unit as ten of the next smaller unit • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

How many tens are equal to 1 hundred?

Tens: _____

True or false?

3 hundreds 4 tens can be renamed as 2 hundreds 14 tens.

A. True

B. False

Which shows 52 renamed so that you can take away 8 ones?

- A.** 6 tens 2 ones
- B.** 4 tens 12 ones
- C.** 4 tens 2 ones
- D.** 5 tens 12 ones

Rename 4 thousands as hundreds.

Hundreds: _____

True or false?

7 tens 3 ones = 6 tens 3 ones

A. True

B. False

SET 62 • 5.NBT.B.7

FORWARD 1

Add and subtract multi-digit decimals with the standard algorithm • Grade 6 •
The Number System • 6.NS.B.3

Add.

$$36.075 + 8.9$$

Sum: _____

Subtract.

$$41.2 - 7.865$$

Difference: _____

Add.

$$0.68 + 4.795$$

- A.** 4.863
- B.** 11.475
- C.** 5.375
- D.** 5.475

Subtract.

$$9 - 2.468$$

Difference: _____

True or false?

$$12.5 + 0.375 = 12.875$$

A. True

B. False

SET 62 • 5.NBT.B.7

FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-3.25 + 7.6$$

Value: _____

Find the value.

$$0.8 - 2.35$$

Value: _____

Find the value.

$$-1.75 + (-0.6)$$

- A.** 2.35
- B.** 1.15
- C.** -2.35
- D.** -1.15

True or false?

$$5.5 - 7.25 = -1.75$$

A. True

B. False

Find the value.

$$-9.4 - (-2.65)$$

Value: _____

Multiply decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiplying whole numbers	4.NBT.B.5
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
BACKWARD 4	Multiply a whole number by tenths or hundredths	4.NF.B.4.b
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 20 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$3.4 \times 6$$

Product: _____

Multiply.

$$0.7 \times 0.5$$

Product: _____

Multiply.

$$2.5 \times 1.2$$

A. 3

B. 30

C. 3.7

D. 0.3

Multiply.

$$0.06 \times 8$$

Product: _____

A pen costs \$1.25.

How much do 7 pens cost?

Cost: _____

SET 20 • 5.NBT.B.7

BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 6$$

Product: _____

Multiply.

$$25 \times 12$$

Product: _____

True or false?

$$125 \times 7 = 785$$

A. True

B. False

$$18 \times 4 = ?$$

A. 72

B. 82

C. 48

D. 64

Multiply.

$$106 \times 8$$

Product: _____

SET 20 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: _____

True or false?

0.4 is 4 tenths.

A. True

B. False

Which decimal is 204 tenths?

- A.** 204
- B.** 20.4
- C.** 2.04
- D.** 0.204

How many hundredths are in 0.48?

Hundredths: _____

True or false?

0.06 is 6 tenths.

A. True

B. False

SET 20 • 5.NBT.B.7

BACKWARD 4

Multiply a whole number by tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$4 \times \frac{3}{10}$$

Product: _____

True or false?

$$6 \times \frac{5}{100} = \frac{30}{100}$$

A. True

B. False

$$3 \times \frac{7}{10} = ?$$

A. $\frac{7}{30}$

B. $\frac{21}{10}$

C. $\frac{10}{10}$

D. $\frac{21}{30}$

A cup holds $\frac{3}{10}$ liter.

How many liters do 5 cups hold? Write a fraction.

Liters: _____

True or false?

$$8 \times \frac{2}{100} = \frac{16}{800}$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.45 \times 0.6$$

Product: _____

Multiply.

$$12.08 \times 2.5$$

Product: _____

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

Multiply.

$$7.2 \times 1.15$$

Product: _____

True or false?

$$0.03 \times 0.03 = 0.09$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-3.4 \times 6$$

Product: _____

Find the product.

$$(-0.7) \times (-0.5)$$

Product: _____

Find the product.

$$2.5 \times (-1.2)$$

A. -0.3

B. -30

C. 3

D. -3

Find the product.

$$(-0.06) \times 8$$

Product: _____

True or false?

$$(-1.25) \times (-4) = -5$$

A. True

B. False

Use models and place value to multiply decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiply a fraction in tenths or hundredths by a whole number	4.NF.B.4.b
BACKWARD 2	Multiplication as equal groups	3.OA.A.1
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 63 • 5.NBT.B.7

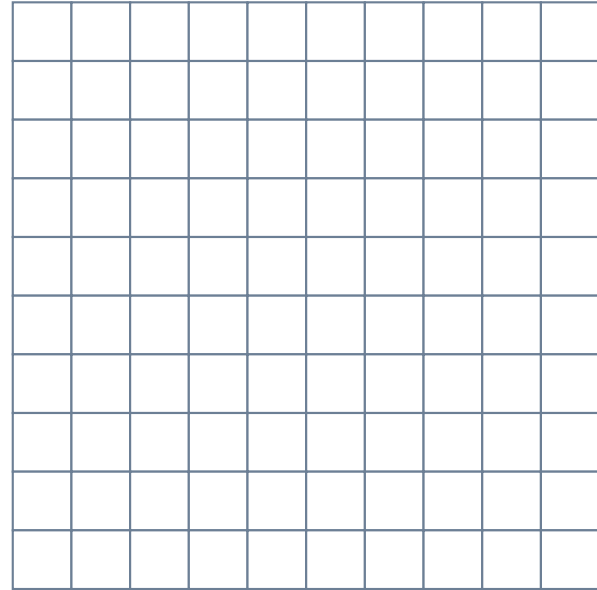
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show
 3×0.15 .

Then write the product.

Product: _____



On a hundredths grid, 0.3×0.4 is where 3 tenths (3 columns) and 4 tenths (4 rows) overlap.

Explain why the product is 0.12.

Explanation: _____

Which explains why $4 \times 0.25 = 1$?

- A.** 4 groups of 25 hundredths is 100 hundredths, which is 1 whole.
- B.** $4 + 0.25$ is about 4, so the product is about 1.
- C.** 0.25 has two decimal places, so the product is 0.01.
- D.** $4 \times 25 = 100$, so the product is 100.

True or false?

$2 \times 0.35 = 0.70$ because 2 groups of 35 hundredths is 70 hundredths.

A. True

B. False

Lia says $0.5 \times 0.6 = 3.0$.

Give the correct product and use place value to explain her mistake.

Correct product: _____

Explanation: _____

SET 63 • 5.NBT.B.7

BACKWARD 1

Multiply a fraction in tenths or hundredths by a whole number • Grade 4 •
Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$3 \times \frac{15}{100}$$

Product: _____

True or false?

$$2 \times \frac{35}{100} = \frac{70}{100}$$

A. True

B. False

$$4 \times \frac{6}{10} = ?$$

A. $\frac{6}{40}$

B. $\frac{24}{40}$

C. $\frac{24}{10}$

D. $\frac{10}{10}$

Write $7 \times \frac{2}{10}$ as one fraction.

Fraction: _____

True or false?

$$6 \times \frac{4}{100} = \frac{24}{600}$$

A. True

B. False

SET 63 • 5.NBT.B.7

BACKWARD 2

Multiplication as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

There are 3 groups with 15 counters in each group.
How many counters are there in all?

Counters: _____

True or false?

4×25 can mean 4 groups of 25.

A. True

B. False

Which can 2×35 mean?

- A.** 2 more than 35
- B.** 2 groups of 3 and 5
- C.** 35 take away 2
- D.** 2 groups of 35

Write a multiplication equation for 6 groups of 4.

Equation: _____

True or false?

3×8 means 3 more than 8.

A. True

B. False

SET 63 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$0.24 \times 0.5$$

Product: _____

Multiply.

$$3.6 \times 0.15$$

Product: _____

Multiply.

$$1.6 \times 0.45$$

A. 0.072

B. 0.72

C. 2.05

D. 7.2

Multiply.

$$0.08 \times 0.7$$

Product: _____

True or false?

$$2.5 \times 0.04 = 1$$

A. True

B. False

SET 63 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-0.3) \times 0.4$$

Product: _____

Find the product.

$$(-2.5) \times (-0.6)$$

Product: _____

Find the product.

$$3 \times (-0.15)$$

A. -0.45

B. 0.45

C. -0.045

D. -4.5

True or false?

$$(-0.2) \times (-0.2) \times (-0.2) = -0.008$$

A. True

B. False

Find the product.

$$-1.2 \times 0.05$$

Product: _____

Divide decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Dividing whole numbers	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 4	Rename decimals as tenths or hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 21 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Divide.

$$8.4 \div 4$$

Quotient: _____

Divide.

$$3.6 \div 0.6$$

Quotient: _____

Divide.

$$0.45 \div 0.05$$

A. 0.09

B. 0.9

C. 90

D. 9

Divide.

$$7.5 \div 0.3$$

Quotient: _____

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: _____

SET 21 • 5.NBT.B.7

BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$84 \div 4$$

Quotient: _____

Divide.

$$360 \div 6$$

Quotient: _____

True or false?

$$450 \div 5 = 9$$

A. True

B. False

$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

Divide.

$$48 \div 6$$

Quotient: _____

SET 21 • 5.NBT.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 6$ by thinking $6 \times \square = 36$.

Quotient: _____

True or false?

$7 \times 9 = 63$, so $63 \div 7 = 9$.

A. True

B. False

Which fact helps you find $56 \div 8$?

A. $56 \times 8 = 448$

B. $8 \times 7 = 56$

C. $8 + 48 = 56$

D. $8 - 7 = 1$

What number goes in the box?

$$\square \times 5 = 45$$

Number: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 21 • 5.NBT.B.7

BACKWARD 4

Rename decimals as tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 3.6?

Tenths: _____

True or false?

0.45 is 45 hundredths.

A. True

B. False

How many hundredths are in 0.05?

A. 500

B. 0.5

C. 5

D. 50

How many tenths are in 0.6?

Tenths: _____

True or false?

7.5 is 75 hundredths.

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: _____

Divide.

$$9.072 \div 2.4$$

Quotient: _____

Divide.

$$0.081 \div 0.09$$

A. 90

B. 0.09

C. 0.9

D. 9

Divide.

$$52.5 \div 1.25$$

Quotient: _____

True or false?

$$1.8 \div 0.006 = 3$$

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: _____

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: _____

Find the quotient.

$$0.45 \div (-0.05)$$

A. -9

B. -0.9

C. 0.9

D. 9

Find the quotient.

$$-7.5 \div 0.3$$

Quotient: _____

True or false?

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

A. True

B. False

Use models and place value to divide decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Division as the number of equal groups	3.OA.A.2
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 64 • 5.NBT.B.7

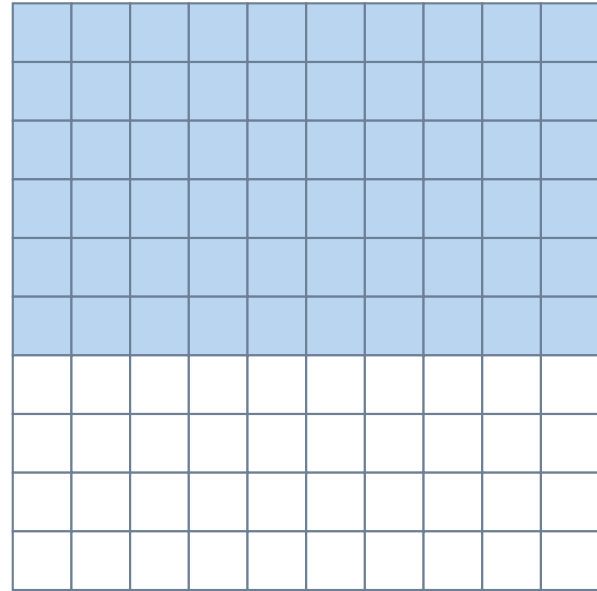
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.6 shaded.

Circle groups of 0.15 to find $0.6 \div 0.15$. Then write the quotient.

Quotient: _____



Which explains why $1.8 \div 0.3 = 6$?

- A.** 0.3 has one decimal place, so the quotient is 0.6.
- B.** There are 6 groups of 1.8 in 0.3.
- C.** $1.8 - 0.3 = 6$
- D.** There are 6 groups of 3 tenths in 18 tenths.

Explain how to find $2.4 \div 4$ by thinking of 2.4 as 24 tenths.

Quotient: _____

Explanation: _____

True or false?

$4.5 \div 0.9 = 5$ because there are 5 groups of 9 tenths in 45 tenths.

A. True

B. False

Sam says $0.8 \div 0.2 = 0.4$.

Give the correct quotient and use a model or place value to explain why he is wrong.

Correct quotient: _____

Explanation: _____

SET 64 • 5.NBT.B.7

BACKWARD 1

Division as the number of equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

18 counters are put into groups of 3.

How many groups are there?

Groups: _____

True or false?

$24 \div 4$ can mean "how many groups of 4 are in 24?"

A. True

B. False

Which question does $35 \div 5$ answer?

- A.** What is 35 take away 5?
- B.** What is 35 more than 5?
- C.** How many groups of 5 are in 35?
- D.** How many are in 5 groups of 35?

There are 28 apples. Each bag holds 7 apples.

How many bags are needed?

Bags: _____

True or false?

$12 \div 3$ means 12 groups of 3.

A. True

B. False

SET 64 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 1.8?

Tenths: _____

True or false?

0.15 is 15 hundredths.

A. True

B. False

How many hundredths are in 0.6?

- A.** 60
- B.** 0.6
- C.** 600
- D.** 6

Write 45 tenths as a decimal.

Decimal: _____

True or false?

2.4 is 24 hundredths.

A. True

B. False

SET 64 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$3.75 \div 0.25$$

Quotient: _____

Divide.

$$0.096 \div 0.12$$

Quotient: _____

Divide.

$$8.1 \div 0.09$$

A. 9

B. 90

C. 0.9

D. 900

Divide.

$$14.4 \div 1.2$$

Quotient: _____

True or false?

$$0.5 \div 0.02 = 2.5$$

A. True

B. False

SET 64 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-1.8 \div 0.3$$

Quotient: _____

Find the quotient.

$$(-0.6) \div (-0.15)$$

Quotient: _____

Find the quotient.

$$2.4 \div (-4)$$

A. 0.6

B. -6

C. -0.06

D. -0.6

True or false?

$$(-4.5) \div (-0.9) = -5$$

A. True

B. False

Find the quotient.

$$-0.08 \div 0.2$$

Quotient: _____

Solve real-world problems with decimal operations

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Word problems with money and measurement	4.MD.A.2
BACKWARD 2	Hundredths on a grid	4.NF.C.6
FORWARD 1	Real-world problems with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 22 • 5.NBT.B.7

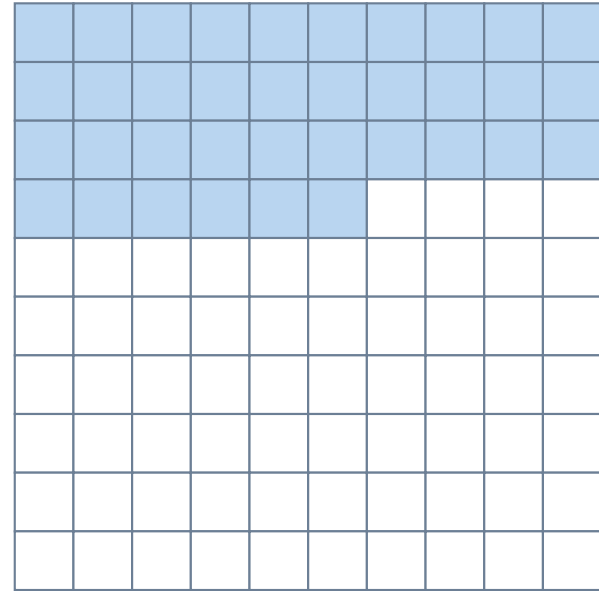
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: _____



Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: _____

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound
- B.** 0.5 pound
- C.** 2 pounds
- D.** 12.5 pounds

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: _____

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: _____

SET 22 • 5.NBT.B.7

BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: _____

A board is 120 cm long. It is cut into 4 equal pieces.
How long is each piece?

Length: _____

True or false?

Two quarters and three dimes are worth 65 cents.

A. True

B. False

Ann has \$5. She spends \$2.25.

How much money is left?

A. \$7.25

B. \$3.25

C. \$2.75

D. \$2.25

A jug holds 2 liters.

How many milliliters is that?

Milliliters: _____

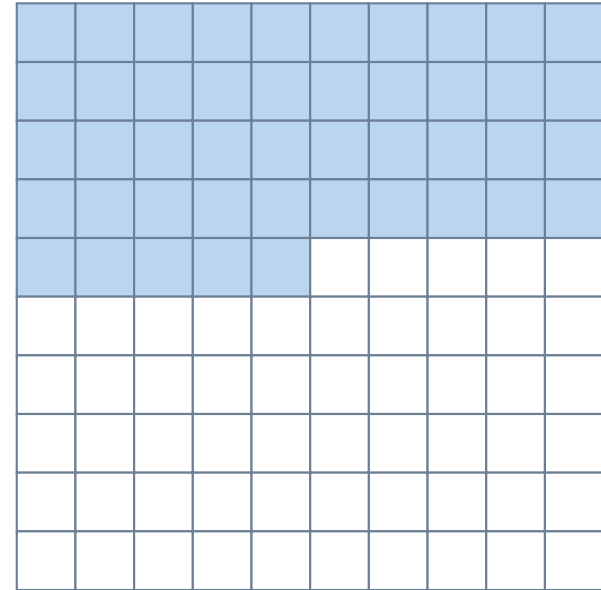
SET 22 • 5.NBT.B.7

BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal does the shaded part of the grid show?

Decimal: _____

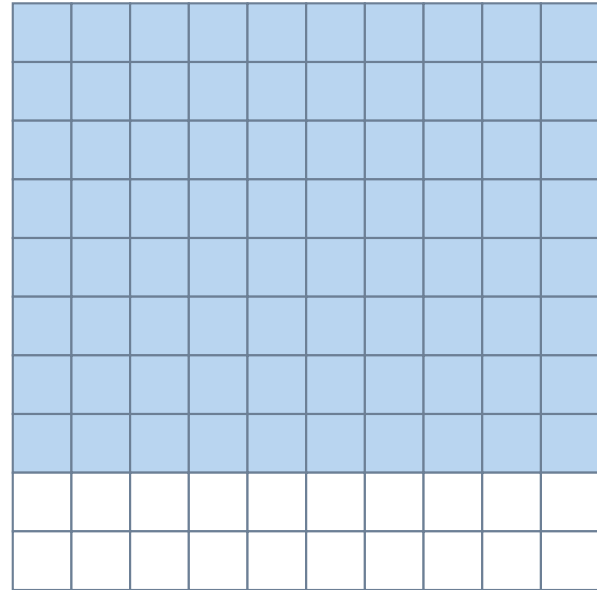


True or false?

The shaded part of the grid shows 0.8.

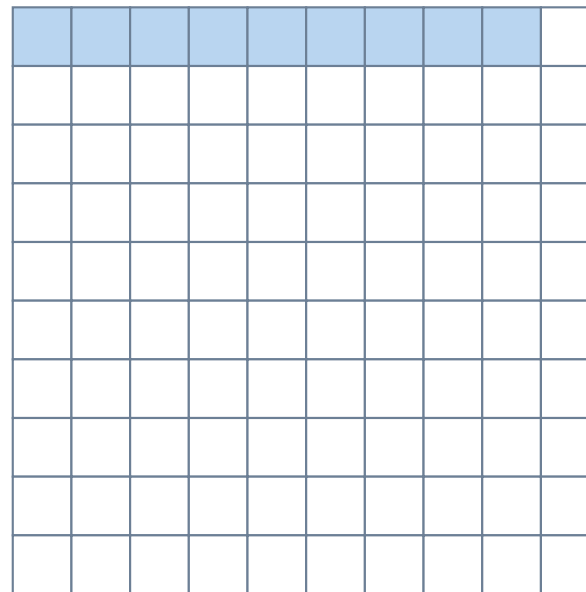
A. True

B. False



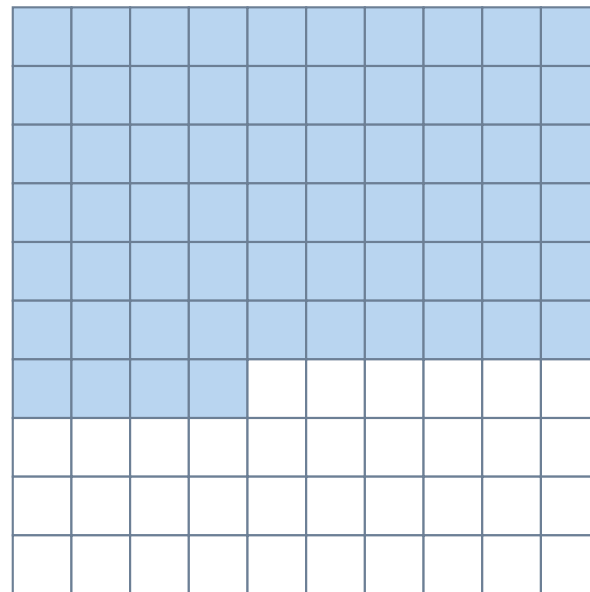
Which decimal does the shaded part of the grid show?

- A.** 0.9
- B.** 9
- C.** 0.009
- D.** 0.09



How many more squares
must be shaded to fill the
whole grid?

Squares: _____



True or false?

0.5 of a hundredths grid is 5 squares.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: _____

Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A.** \$3.00
- B.** \$30.00
- C.** \$2.70
- D.** \$3.70

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: _____

True or false?

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

- A. True
- B. False

SET 22 • 5.NBT.B.7

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A submarine at -45.5 m rises 12.75 m and then dives 20 m.

What is its final position?

Position: _____

A store's profit was $-\$125.40$ in May and $\$310.15$ in June.

What was the total profit for the two months?

Total: _____

The temperature was 4°F . It dropped 1.5°F each hour for 6 hours.

What was the final temperature?

A. -5°F

B. 13°F

C. 5°F

D. -9°F

An account changes by $-\$2.40$ each week for 5 weeks.
What is the total change?

Total change: _____

True or false?

$$-3.6 \div 0.4 = -0.9$$

A. True

B. False

Add fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Adding fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Find the least common multiple	6.NS.B.4
FORWARD 2	Add rational numbers	7.NS.A.1.d

SET 23 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: _____

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A. $1\frac{5}{24}$

B. $1\frac{1}{24}$

C. $\frac{8}{24}$

D. $\frac{8}{14}$

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: _____

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: _____

SET 23 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{4} = \frac{4}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{3}{5}$?

A. $\frac{6}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{10}$

D. $\frac{6}{10}$

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: _____

True or false?

$$\frac{2}{7} = \frac{4}{21}$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

A. True

B. False

$$\frac{7}{10} + \frac{6}{10} = ?$$

A. $1\frac{6}{10}$

B. $1\frac{3}{10}$

C. $\frac{13}{20}$

D. $\frac{1}{10}$

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{5} = 1$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five positive multiples of 4.

Multiples: _____

True or false?

10 is a multiple of both 3 and 4.

A. True

B. False

Which number is a multiple of both 6 and 8?

A. 16

B. 30

C. 24

D. 14

Name a positive number that is a multiple of both 5 and 10.

Number: _____

True or false?

18 is a multiple of both 4 and 6.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 1

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 10.

LCM: _____

Find the least common multiple of 8 and 12.

LCM: _____

What is the least common multiple of 4 and 9?

- A.** 72
- B.** 13
- C.** 18
- D.** 36

Find the least common multiple of 5 and 15.

LCM: _____

True or false?

The least common multiple of 6 and 9 is 54.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{2}{3} + \frac{1}{4}$$

Value: _____

Find the value.

$$\frac{3}{5} + \left(-\frac{7}{10}\right)$$

Value: _____

Find the value.

$$-\frac{1}{2} + \left(-\frac{3}{8}\right)$$

A. $-\frac{7}{8}$

B. $-\frac{4}{10}$

C. $\frac{1}{8}$

D. $-\frac{1}{8}$

Find the value.

$$-1\frac{1}{3} + \frac{5}{6}$$

Value: _____

True or false?

$$-\frac{3}{4} + \frac{3}{4} = 0$$

A. True

B. False

Subtract fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Subtracting fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 24 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

Subtract.

$$\frac{7}{8} - \frac{1}{2}$$

Difference: _____

Subtract.

$$\frac{4}{5} - \frac{2}{3}$$

A. $\frac{2}{2}$

B. $\frac{1}{5}$

C. $\frac{6}{15}$

D. $\frac{2}{15}$

Subtract.

$$\frac{9}{10} - \frac{2}{5}$$

Difference: _____

Subtract.

$$1 - \frac{3}{7}$$

Difference: _____

SET 24 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{3}$?

A. $\frac{10}{15}$

B. $\frac{4}{9}$

C. $\frac{3}{4}$

D. $\frac{2}{6}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{3}{4} = \frac{9}{16}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 2

Subtracting fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Subtract.

$$\frac{7}{9} - \frac{2}{9}$$

Difference: _____

True or false?

$$\frac{5}{8} - \frac{3}{8} = \frac{2}{8}$$

A. True

B. False

$$\frac{11}{12} - \frac{4}{12} = ?$$

A. $\frac{7}{24}$

B. $\frac{7}{12}$

C. $\frac{15}{12}$

D. $\frac{4}{12}$

Subtract.

$$1 - \frac{3}{5}$$

Difference: _____

True or false?

$$\frac{9}{10} - \frac{4}{10} = \frac{5}{20}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four positive multiples of 6.

Multiples: _____

True or false?

15 is a multiple of both 3 and 5.

A. True

B. False

Which number is a multiple of both 4 and 5?

A. 15

B. 20

C. 9

D. 25

Name a positive number that is a multiple of both 3 and 8.

Number: _____

True or false?

16 is a multiple of both 4 and 6.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{3} = \frac{5}{6}$$

x = _____

Solve for y.

$$y - \frac{2}{5} = \frac{3}{10}$$

$$y = \underline{\hspace{10cm}}$$

Solve for n.

$$\frac{3}{4} + n = 1\frac{1}{2}$$

A. $1\frac{1}{4}$

B. $\frac{3}{4}$

C. $2\frac{1}{4}$

D. $\frac{1}{4}$

Solve for m.

$$m + \frac{5}{8} = 2$$

m = _____

True or false?

The solution of $x - \frac{1}{6} = \frac{2}{3}$ is $x = \frac{5}{6}$.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$\frac{1}{4} - \frac{5}{6}$$

Value: _____

Find the value.

$$-\frac{2}{3} - \frac{1}{6}$$

Value: _____

Find the value.

$$\frac{1}{2} - \left(-\frac{3}{4}\right)$$

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. $1\frac{1}{4}$

D. $-1\frac{1}{4}$

Find the value.

$$-\frac{3}{5} - (-\frac{1}{2})$$

Value: _____

True or false?

$$\frac{2}{9} - \frac{5}{9} = \frac{1}{3}$$

A. True

B. False

Add mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Adding mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Mixed numbers and fractions greater than 1	4.NF.B.3.b
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Solve equations of the form $px + q = r$ with fractions	7.EE.B.4.a

SET 25 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$2\frac{1}{3} + 1\frac{1}{2}$$

Sum: _____

Add.

$$4\frac{3}{4} + 2\frac{1}{6}$$

Sum: _____

Add.

$$1\frac{2}{5} + 3\frac{1}{2}$$

A. $4\frac{3}{7}$

B. $4\frac{3}{10}$

C. $5\frac{1}{10}$

D. $4\frac{9}{10}$

Add.

$$3\frac{5}{8} + 2\frac{3}{4}$$

Sum: _____

Add.

$$5\frac{2}{3} + 1\frac{5}{9}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 1

Adding mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Add.

$$1\frac{2}{5} + 2\frac{1}{5}$$

Sum: _____

Add.

$$4\frac{1}{6} + 1\frac{4}{6}$$

Sum: _____

True or false?

$$2\frac{3}{4} + 1\frac{3}{4} = 3\frac{6}{8}$$

A. True

B. False

$$2\frac{2}{3} + 1\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $4\frac{1}{3}$

D. $4\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.c • Set 25 • BACKWARD 1 • Q4/5

Add.

$$3\frac{7}{8} + \frac{3}{8}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 2

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{17}{5}$ as a mixed number.

Mixed number: _____

True or false?

$$1\frac{1}{3} = \frac{4}{3}$$

A. True

B. False

Which fraction is equal to $3\frac{1}{2}$?

A. $\frac{31}{2}$

B. $\frac{7}{2}$

C. $\frac{5}{2}$

D. $\frac{6}{2}$

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

SET 25 • 5.NF.A.1

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{2}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{1}{2}$?

A. $\frac{5}{10}$

B. $\frac{2}{3}$

C. $\frac{2}{2}$

D. $\frac{1}{4}$

Find the missing number.

$$\frac{2}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{5}{6} = \frac{10}{18}$$

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 2\frac{1}{2} = 6\frac{1}{4}$$

x = _____

Solve for y.

$$y - 1\frac{2}{3} = 3\frac{1}{6}$$

y = _____

Solve for n.

$$n + 4\frac{3}{8} = 7$$

A. $3\frac{5}{8}$

B. $2\frac{3}{8}$

C. $11\frac{3}{8}$

D. $2\frac{5}{8}$

A board was $8\frac{1}{2}$ feet long. After a piece was cut off, it was $5\frac{3}{4}$ feet long.

Write and solve an equation to find the length p of the piece.

Equation and solution: _____

True or false?

The solution of $x - 1\frac{1}{4} = 2\frac{1}{2}$ is $x = 1\frac{1}{4}$.

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 2

Solve equations of the form $px + q = r$ with fractions • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$2x + \frac{1}{2} = 3\frac{1}{2}$$

x = _____

Solve for x.

$$\frac{1}{3}x - 1\frac{1}{4} = 2\frac{3}{4}$$

x = _____

Solve for x.

$$4x - 2\frac{1}{3} = 1$$

A. $1\frac{1}{12}$

B. $\frac{1}{6}$

C. $\frac{5}{6}$

D. $\frac{1}{3}$

Jo bought 3 bags of flour that weigh the same and one $1\frac{1}{2}$ -pound bag of sugar. The total weight was $9\frac{3}{4}$ pounds.

How much does each bag of flour weigh?

Weight: _____

True or false?

The solution of $\frac{1}{2}(x + 3) = 4$ is $x = 5$.

A. True

B. False

Subtract mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Subtracting mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Rename a mixed number by decomposing a whole	4.NF.B.3.b
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 65 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$5\frac{3}{4} - 2\frac{1}{6}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{1}{2}$$

A. $3\frac{3}{10}$

B. $2\frac{7}{10}$

C. $3\frac{7}{10}$

D. $2\frac{3}{10}$

Subtract.

$$6 - 2\frac{2}{3}$$

Difference: _____

Subtract.

$$7\frac{1}{4} - 3\frac{5}{6}$$

Difference: _____

True or false?

$$3\frac{1}{2} - 1\frac{2}{3} = 1\frac{5}{6}$$

A. True

B. False

SET 65 • 5.NF.A.1

BACKWARD 1

Subtracting mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$4\frac{5}{6} - 1\frac{4}{6}$$

Difference: _____

$$5\frac{1}{3} - 2\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $2\frac{2}{3}$

D. $2\frac{1}{3}$

Subtract.

$$6\frac{1}{4} - 3\frac{3}{4}$$

Difference: _____

True or false?

$$7\frac{2}{5} - 4\frac{4}{5} = 3\frac{2}{5}$$

A. True

B. False

Subtract.

$$5 - 1\frac{3}{8}$$

Difference: _____

SET 65 • 5.NF.A.1

BACKWARD 2

Rename a mixed number by decomposing a whole • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Fill in the blank.

$$3\frac{1}{4} = 2\frac{?}{4}$$

Blank: _____

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

Which is equal to $4\frac{1}{8}$?

A. $4\frac{9}{8}$

B. $5\frac{1}{8}$

C. $3\frac{1}{8}$

D. $3\frac{9}{8}$

Fill in the blank.

$$6 = 5\frac{?}{5}$$

Blank: _____

True or false?

$$2\frac{3}{10} = 1\frac{3}{10}$$

A. True

B. False

SET 65 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 3\frac{2}{3} = 7\frac{1}{4}$$

x = _____

Solve for y.

$$9\frac{1}{2} - y = 4\frac{5}{6}$$

y = _____

Solve for n.

$$n + 1\frac{5}{8} = 4\frac{1}{4}$$

A. $2\frac{5}{8}$

B. $2\frac{3}{8}$

C. $3\frac{5}{8}$

D. $5\frac{7}{8}$

A tank had $10\frac{1}{4}$ gallons of water. After some water was used, $6\frac{2}{3}$ gallons were left.

Write and solve an equation to find the amount w used.

Equation and solution: _____

True or false?

The solution of $x + 2\frac{3}{4} = 5$ is $x = 2\frac{1}{4}$.

A. True

B. False

SET 65 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$2\frac{1}{4} - 5\frac{1}{2}$$

Value: _____

Find the value.

$$-1\frac{2}{3} - 2\frac{1}{6}$$

Value: _____

Find the value.

$$3\frac{1}{2} - (-1\frac{3}{4})$$

A. $1\frac{3}{4}$

B. $-1\frac{3}{4}$

C. $5\frac{1}{4}$

D. $-5\frac{1}{4}$

Find the value.

$$-4\frac{1}{3} - (-1\frac{5}{6})$$

Value: _____

True or false?

$$1\frac{1}{4} - 3 = -1\frac{3}{4}$$

A. True

B. False

Solve word problems by adding and subtracting fractions

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Word problems with like denominators	4.NF.B.3.d
BACKWARD 2	Choosing addition or subtraction	2.OA.A.1
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Write and solve equations of the form $x + p = q$	6.EE.B.7
FORWARD 2	Write and solve equations of the form $px + q = r$	7.EE.B.4.a

SET 26 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Mia walked $\frac{3}{4}$ mile to school and then $\frac{2}{3}$ mile to the park.

How far did she walk in all?

Distance: _____

A jug had $\frac{7}{8}$ gallon of milk. Ben drank $\frac{1}{3}$ gallon.

How much milk is left?

Milk left: _____

Ana used $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar.

How many cups did she use in all?

A. $4\frac{1}{4}$ cups

B. $3\frac{3}{4}$ cups

C. $4\frac{3}{4}$ cups

D. $3\frac{1}{4}$ cups

A ribbon is $5\frac{1}{6}$ feet long. Lea cuts off $2\frac{1}{2}$ feet.

How long is the ribbon now?

Length: _____

Tom read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of the book on Tuesday.

What fraction of the book does he still have to read?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 1

Word problems with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

Kim ate $\frac{3}{8}$ of a pizza and Joe ate $\frac{2}{8}$ of the pizza.

What fraction of the pizza did they eat in all?

Fraction: _____

A rope is $\frac{9}{10}$ meter long. $\frac{4}{10}$ meter is cut off.

How much rope is left?

Rope left: _____

True or false?

$\frac{2}{6}$ of a garden is corn and $\frac{3}{6}$ is beans. Corn and beans cover $\frac{5}{6}$ of the garden.

A. True

B. False

Lu had $\frac{7}{12}$ of a cake. She gave away $\frac{5}{12}$ of the cake.

How much of the cake is left?

A. $\frac{12}{12}$

B. $\frac{2}{12}$

C. $\frac{35}{12}$

D. $\frac{2}{24}$

A tank is $\frac{3}{5}$ full. Water is added to fill another $\frac{1}{5}$ of the tank.

How full is the tank now?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 2

Choosing addition or subtraction • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

A class has 14 boys and 13 girls.

Which operation finds the number of students in all?

- A.** Subtraction
- B.** Division
- C.** Addition
- D.** Multiplication

True or false?

To find how much is left after some is used, you subtract.

A. True

B. False

Sam had 42 marbles. He lost 15 marbles.
How many marbles are left?

Marbles: _____

A shelf holds 35 books. 18 books are on it now.

Which equation finds how many more books fit?

A. $35 - 18 = \square$

B. $18 - 35 = \square$

C. $35 \times 18 = \square$

D. $35 + 18 = \square$

Liz has 26 stickers. She gets 17 more.

How many stickers does she have now?

Stickers: _____

SET 26 • 5.NF.A.2

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{2}{3} = \frac{8}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{15}{16}$

B. $\frac{10}{12}$

C. $\frac{5}{12}$

D. $\frac{6}{5}$

Rewrite $\frac{1}{2}$ and $\frac{1}{3}$ as fractions with the denominator 6.

$$\frac{1}{2} = \underline{\hspace{10cm}}$$

$$\frac{1}{3} = \underline{\hspace{10cm}}$$

True or false?

$$\frac{3}{8} = \frac{6}{24}$$

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 1

Write and solve equations of the form $x + p = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

After using $\frac{2}{3}$ cup of oil, Ray has $\frac{3}{4}$ cup left.

Write and solve an equation to find how much oil c he started with.

Equation and solution: _____

A plant grew $\frac{5}{8}$ inch and is now $4\frac{1}{4}$ inches tall.

Write and solve an equation to find its starting height h .

Equation and solution: _____

$\frac{3}{10}$ inch of rain fell on day 1. In all, $\frac{9}{10}$ inch fell on days 1 and 2.

Which equation finds the rain r on day 2?

A. $\frac{3}{10}r = \frac{9}{10}$

B. $r + \frac{9}{10} = \frac{3}{10}$

C. $\frac{3}{10} + r = \frac{9}{10}$

D. $r - \frac{3}{10} = \frac{9}{10}$

Solve for x.

$$x + \frac{2}{9} = \frac{2}{3}$$

x = _____

True or false?

If $n - \frac{1}{4} = \frac{5}{8}$, then $n = \frac{7}{8}$.

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 2

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A recipe uses $\frac{3}{4}$ cup of sugar per batch, plus $\frac{1}{2}$ cup for frosting. Ali used $2\frac{3}{4}$ cups of sugar in all.

How many batches did he make?

Batches: _____

Solve for x.

$$\frac{2}{3}x + \frac{1}{6} = \frac{5}{6}$$

x = _____

A 6-foot board is cut into 4 equal pieces, with $\frac{1}{2}$ foot left over.

How long is each piece?

A. $1\frac{1}{2}$ feet

B. $1\frac{5}{8}$ feet

C. $1\frac{3}{8}$ feet

D. $5\frac{1}{2}$ feet

Solve for x.

$$5x - \frac{3}{4} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $2x + \frac{1}{3} = 1$ is $x = \frac{2}{3}$.

A. True

B. False

Use benchmark fractions to estimate and to check answers

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Comparing fractions to $\frac{1}{2}$ and 1	4.NF.A.2
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Reason about the size of a quotient of fractions	6.NS.A.1
FORWARD 2	Use estimation to judge whether an answer is reasonable	7.EE.B.3

SET 27 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

Ella says $\frac{3}{4} + \frac{1}{6} = \frac{4}{10}$. Her answer must be wrong, because $\frac{4}{10}$ is less than $\frac{3}{4}$, one of the fractions she added.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{12}$?

- A.** About $\frac{1}{2}$
- B.** About $2\frac{1}{2}$
- C.** About $1\frac{1}{2}$
- D.** About 3

Is $\frac{3}{5} + \frac{4}{9}$ more or less than 1?

Use benchmark fractions to explain.

Answer: _____

Explanation: _____

Jen ran $\frac{9}{10}$ mile and walked $\frac{1}{8}$ mile.

Estimate the total distance to the nearest whole mile.

Estimate: _____

Leo says $\frac{5}{6} - \frac{1}{3} = \frac{4}{3}$.

Which statement explains why his answer is not reasonable?

- A.** The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.
- B.** The denominators are different.
- C.** $\frac{4}{3}$ is less than 1.
- D.** The answer should be 0.

SET 27 • 5.NF.A.2

BACKWARD 1

Comparing fractions to $\frac{1}{2}$ and 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{5}{8}$ is less than $\frac{1}{2}$.

A. True

B. False

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{3}{8}$

C. $\frac{2}{5}$

D. $\frac{4}{7}$

Is $\frac{5}{9}$ closest to 0, $\frac{1}{2}$, or 1?

Answer: _____

True or false?

$\frac{6}{10}$ is less than $\frac{1}{2}$.

A. True

B. False

Is $\frac{11}{12}$ closer to $\frac{1}{2}$ or to 1?

Answer: _____

SET 27 • 5.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

Which point is at $\frac{1}{3}$?



A. Point A

B. Point B

C. Point C

D. Point D

What fraction is at point B?



B = _____

True or false?

$\frac{7}{8}$ is closer to 1 than to $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 1

Reason about the size of a quotient of fractions • Grade 6 • The Number System • 6.NS.A.1

True or false?

$\frac{3}{4} \div \frac{1}{8}$ is greater than 1.

A. True

B. False

Without dividing, which quotient is less than 1?

A. $\frac{3}{4} \div \frac{1}{2}$

B. $\frac{1}{3} \div \frac{1}{2}$

C. $2 \div \frac{1}{2}$

D. $\frac{5}{6} \div \frac{1}{3}$

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

Quotient: _____

Estimate $2\frac{7}{8} \div \frac{1}{2}$ by first rounding $2\frac{7}{8}$ to the nearest whole number.

Estimate: _____

True or false?

$\frac{1}{2} \div 4$ is greater than $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A board $9\frac{7}{8}$ feet long is cut into 3 equal pieces.

Round the length of the board to the nearest whole foot.
Then use it to estimate the length of each piece.

Rounded length: _____

Estimate: _____

Ann earns \$11.85 per hour and works $7\frac{3}{4}$ hours.

Which is the best estimate of her pay?

- A.** \$96
- B.** \$84
- C.** \$120
- D.** \$77

True or false?

$\frac{3}{4}$ of \$79.95 is about \$60.

A. True

B. False

Kai says $4\frac{5}{6} + 3\frac{1}{8}$ is about 9.

Round each mixed number to the nearest whole number and estimate the sum. Is the exact sum more or less than 9?

Your estimate: _____

More or less than 9? _____

A recipe needs $2\frac{1}{3}$ cups of flour per batch.

Round $2\frac{1}{3}$ to the nearest whole number. Then estimate the flour needed for 4 batches.

Rounded amount: _____

Estimate: _____

Interpret a fraction as division of the numerator by the denominator

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	Division with remainders	4.NBT.B.6
BACKWARD 3	Mixed numbers and fractions greater than 1	4.NF.B.3.b
FORWARD 1	Unit rates written as fractions	6.RP.A.2
FORWARD 2	Write fractions as decimals using division	7.NS.A.2.d

SET 28 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 5$ as a fraction.

Fraction: _____

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{4}$?

A. $7 \div 4$

B. $7 - 4$

C. $4 \div 7$

D. 7×4

Write $\frac{13}{6}$ as a division expression. Then write the quotient as a mixed number.

Division: _____

Mixed number: _____

True or false?

When 2 pounds of rice are shared equally among 3 bags, each bag gets $\frac{2}{3}$ pound.

- A. True
- B. False

SET 28 • 5.NF.B.3

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ is one of 4 equal parts of a whole.

A. True

B. False

A sandwich is cut into 3 equal parts.

What fraction of the sandwich is one part?

Fraction: _____

What fraction of the bar is shaded?



A. $\frac{8}{3}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{3}{5}$

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 2

Division with remainders • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide. Write the quotient and the remainder.

$$17 \div 5$$

Quotient and remainder: _____

True or false?

$$23 \div 4 = 6 \text{ R } 1$$

A. True

B. False

$$29 \div 6 = ?$$

A. 5 R 5

B. 5 R 1

C. 4 R 3

D. 4 R 5

Divide. Write the quotient and the remainder.

$$50 \div 8$$

Quotient and remainder: _____

True or false?

$$15 \div 2 = 7 \text{ R } 2$$

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 3

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{9}{4}$ as a mixed number.

Mixed number: _____

Write $1\frac{2}{3}$ as a fraction.

Fraction: _____

True or false?

$$\frac{11}{5} = 1\frac{1}{5}$$

A. True

B. False

Which mixed number is equal to $\frac{7}{3}$?

A. $2\frac{2}{3}$

B. $3\frac{1}{3}$

C. $2\frac{1}{3}$

D. $1\frac{1}{3}$

Write $\frac{15}{4}$ as a mixed number.

Mixed number: _____

SET 28 • 5.NF.B.3

FORWARD 1

Unit rates written as fractions • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A recipe uses 3 cups of flour for every 4 cups of water.
How many cups of flour are used per cup of water?

Cups of flour: _____

8 apples cost \$6.

What is the cost per apple?

Cost per apple: _____

A car goes 150 miles on 4 gallons of gas.
What is the rate in miles per gallon?

- A.** $\frac{4}{150}$
- B.** 600
- C.** 154
- D.** 37.5

There are 5 cups of juice for every 2 pitchers.

How many cups of juice are there per pitcher?

Cups per pitcher: _____

True or false?

The ratio 3 : 8 has the unit rate $\frac{3}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

FORWARD 2

Write fractions as decimals using division • Grade 7 • The Number System •
7.NS.A.2.d

Write $5 \div 8$ as a decimal.

Decimal: _____

Use long division to write $\frac{7}{12}$ as a decimal. Show your work.

Decimal: _____

Which fraction has a decimal form that repeats?

A. $\frac{7}{20}$

B. $\frac{5}{6}$

C. $\frac{9}{25}$

D. $\frac{3}{8}$

Write $11 \div 4$ as a decimal.

Decimal: _____

True or false?

$$\frac{1}{3} = 0.3$$

A. True

B. False

Solve sharing problems whose answers are fractions or mixed numbers

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	Whole numbers as fractions	3.NF.A.3.c
FORWARD 1	Solve unit rate problems	6.RP.A.3.b
FORWARD 2	Unit rates with fractional quantities	7.RP.A.1

SET 29 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

6 friends share 10 granola bars equally.

How many bars does each friend get? Write a mixed number.

Bars: _____

A 50-pound bag of rice is shared equally among 8 families.

How many pounds does each family get?

Pounds: _____

9 yards of ribbon are cut into 4 equal pieces.

How long is each piece?

A. $4\frac{1}{2}$ yards

B. $\frac{4}{9}$ yard

C. $2\frac{1}{9}$ yards

D. $2\frac{1}{4}$ yards

7 liters of juice are poured equally into 3 jugs.

How many liters are in each jug? Between which two whole numbers is your answer?

Liters: _____

Between: _____

True or false?

When 5 cookies are shared equally by 2 children, each child gets $2\frac{1}{2}$ cookies.

A. True

B. False

SET 29 • 5.NF.B.3

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

24 stickers are shared equally by 6 friends.

How many stickers does each friend get?

Stickers: _____

35 chairs are set out in 5 equal rows.

How many chairs are in each row?

A. 30

B. 6

C. 40

D. 7

True or false?

If 4 boxes hold 28 toys, with the same number in each box, each box holds 7 toys.

A. True

B. False

A baker puts 45 muffins into bags of 9.
How many bags does she fill?

Bags: _____

18 pencils are shared equally by 3 students.

How many pencils does each student get?

Pencils: _____

SET 29 • 5.NF.B.3

BACKWARD 2

Whole numbers as fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

True or false?

$$\frac{6}{3} = 3$$

A. True

B. False

Write 4 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 3?

A. $\frac{3}{3}$

B. $\frac{2}{6}$

C. $\frac{3}{2}$

D. $\frac{6}{2}$

How many halves are in 5 wholes?

Halves: _____

True or false?

$$\frac{8}{8} = 8$$

A. True

B. False

SET 29 • 5.NF.B.3

FORWARD 1

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

3 pounds of grapes cost \$7.

What is the cost per pound?

Cost per pound: _____

A car travels 100 miles in 3 hours at a constant speed.
At this rate, how far does it travel in 6 hours?

Distance: _____

8 cans of soup cost \$6.

At this rate, how much do 12 cans cost?

A. \$16

B. \$8

C. \$9

D. \$4

A runner runs 5 miles in 40 minutes at a constant pace.
At this pace, how long does she take to run 8 miles?

Time: _____

True or false?

If 4 pens cost \$3, then 10 pens cost \$7.00.

- A. True
- B. False

SET 29 • 5.NF.B.3

FORWARD 2

Unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail crawls $\frac{2}{3}$ foot in $\frac{1}{4}$ hour.

What is its speed in feet per hour?

Speed: _____

A recipe uses $\frac{3}{4}$ cup of sugar for every $\frac{1}{2}$ cup of milk.

How many cups of sugar are used per cup of milk?

Cups of sugar: _____

Jo walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour.

What is her speed?

A. 2 miles per hour

B. 3 miles per hour

C. $\frac{1}{3}$ mile per hour

D. $\frac{1}{12}$ mile per hour

A pump fills $\frac{5}{8}$ of a tank in $\frac{1}{2}$ hour.

How much of the tank does it fill per hour?

Tanks per hour: _____

True or false?

$\frac{1}{2}$ cup of juice for every $\frac{1}{4}$ cup of water is a rate of 2 cups of juice per cup of water.

- A. True
- B. False

Multiply a fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A fraction times a whole number as repeated unit fractions	4.NF.B.4.b
BACKWARD 2	Parts of a whole	3.NF.A.1
BACKWARD 3	Splitting a number into equal parts	3.OA.A.2
FORWARD 1	Find a percent of a quantity	6.RP.A.3.c
FORWARD 2	Multistep percent problems	7.RP.A.3

SET 30 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times 12$$

Product: _____

Multiply.

$$5 \times \frac{2}{3}$$

Product: _____

What is $\frac{2}{5}$ of 20?

A. 10

B. 8

C. 4

D. 50

Multiply.

$$\frac{5}{8} \times 6$$

Product: _____

True or false?

$\frac{1}{3} \times 15$ means 15 split into 3 equal parts, with 1 of the parts taken.

- A.** True
- B.** False

SET 30 • 5.NF.B.4.a

BACKWARD 1

A fraction times a whole number as repeated unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Write $\frac{4}{5}$ as a whole number times $\frac{1}{5}$.

Expression: _____

True or false?

$$3 \times \frac{2}{7} = \frac{6}{21}$$

A. True

B. False

$$3 \times \frac{3}{8} = ?$$

A. $3\frac{3}{8}$

B. $\frac{6}{8}$

C. $\frac{9}{8}$

D. $\frac{9}{24}$

Multiply.

$$4 \times \frac{2}{9}$$

Product: _____

True or false?

$$2 \times \frac{3}{5} = \frac{6}{10}$$

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 2

Parts of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ of a bar means 1 of 4 equal parts.

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

What fraction is 2 parts of size $\frac{1}{3}$?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{3}{2}$

How many $\frac{1}{8}$ s are in $\frac{5}{8}$?

Answer: _____

True or false?

$\frac{3}{4}$ means 3 parts of size $\frac{1}{4}$.

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 3

Splitting a number into equal parts • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.2

12 is split into 4 equal parts.

How much is in each part?

Answer: _____

True or false?

20 split into 5 equal parts gives 4 in each part.

A. True

B. False

Which expression shows 18 split into 3 equal parts?

A. $3 \div 18$

B. $18 - 3$

C. $18 \div 3$

D. 18×3

Divide.

$$24 \div 8$$

Quotient: _____

30 is split into 6 equal parts.

How much is in each part?

Answer: _____

SET 30 • 5.NF.B.4.a

FORWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 48?

Answer: _____

What is 60% of 35?

Answer: _____

A \$40 game is 15% off.

How many dollars are taken off the price?

- A.** \$6
- B.** \$15
- C.** \$4
- D.** \$34

A class has 30 students. 70% of them ride the bus.
How many students ride the bus?

Students: _____

True or false?

50% of 18 is 36.

A. True

B. False

SET 30 • 5.NF.B.4.a

FORWARD 2

Multistep percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A \$60 coat is 30% off.

What is the sale price?

Sale price: _____

A meal costs \$25. A 20% tip is added.

What is the total cost?

Total: _____

A \$80 bike has a 6% sales tax added.

What is the total cost?

- A.** \$86.00
- B.** \$4.80
- C.** \$85.60
- D.** \$84.80

A town of 500 people grew by 12%.

What is the new population?

Population: _____

True or false?

A 25% markup on a \$40 item makes the price \$50.

- A.** True
- B.** False

Multiply a fraction by a fraction

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A whole number times a fraction	4.NF.B.4.b
BACKWARD 2	Equivalent fractions and simplest form	4.NF.A.1
FORWARD 1	Divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 31 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{7}$$

Product: _____

Multiply.

$$\frac{3}{4} \times \frac{1}{6}$$

Product: _____

Multiply.

$$\frac{5}{6} \times \frac{3}{10}$$

A. $\frac{15}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{16}$

What is $\frac{1}{2}$ of $\frac{3}{8}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 1

A whole number times a fraction • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$2 \times \frac{3}{10}$$

Product: _____

True or false?

$$5 \times \frac{1}{4} = \frac{5}{20}$$

A. True

B. False

$$4 \times \frac{2}{3} = ?$$

A. $\frac{8}{12}$

B. $\frac{8}{3}$

C. $\frac{6}{3}$

D. $\frac{2}{12}$

Multiply.

$$3 \times \frac{5}{12}$$

Product: _____

True or false?

$$6 \times \frac{1}{6} = 1$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 2

Equivalent fractions and simplest form • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{6}{8}$ as an equivalent fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{10}{15} = \frac{3}{4}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{12}$?

A. $\frac{4}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{1}{3}$

Find the missing number.

$$\frac{9}{12} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{5}{20} = \frac{1}{5}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 1

Divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{4}{5}$$

Quotient: _____

Divide.

$$\frac{3}{4} \div \frac{1}{6}$$

Quotient: _____

Divide.

$$\frac{5}{8} \div \frac{1}{4}$$

A. $\frac{5}{32}$

B. $\frac{1}{2}$

C. $2\frac{1}{2}$

D. $1\frac{1}{4}$

How many $\frac{3}{8}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-\frac{2}{3} \times \frac{3}{4}$$

Product: _____

Find the product.

$$\left(-\frac{4}{5}\right) \times \left(-\frac{5}{6}\right)$$

Product: _____

Find the product.

$$\frac{3}{7} \times \left(-\frac{7}{9}\right)$$

A. $-\frac{3}{7}$

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. $\frac{10}{16}$

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right)$$

Product: _____

True or false?

$$\left(-\frac{2}{5}\right) \times \frac{5}{2} = 1$$

A. True

B. False

Interpret fraction products with tape diagrams and area models

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	Fractions of a whole in a model	3.NF.A.1
BACKWARD 2	Split a whole number into equal parts	3.OA.A.2
FORWARD 1	Use visual models to divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

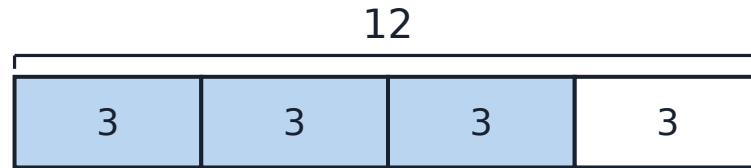
SET 66 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

The tape diagram shows 12 split into 4 equal parts, with 3 parts shaded.

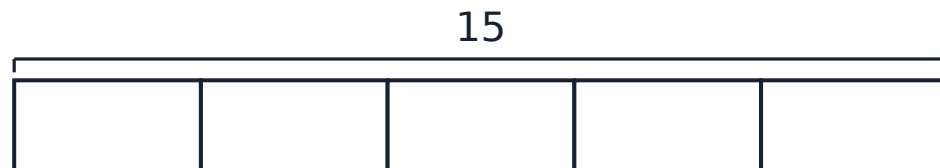
Which product does the shaded part show?



- A.** $\frac{3}{4} \times 12$ **B.** $\frac{4}{3} \times 12$ **C.** $\frac{1}{4} \times 3$ **D.** 3×12

Shade the tape diagram to show $\frac{2}{5} \times 15$.

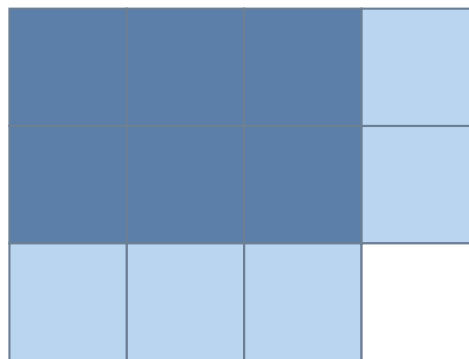
Then write the product.



Product: _____

In the model, the darker part shows a product of two fractions.

Which equation does the model show?



- A.** $\frac{2}{3} + \frac{3}{4} = \frac{5}{7}$ **B.** $\frac{2}{3} \times \frac{3}{4} = \frac{5}{7}$
- C.** $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$ **D.** $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$

Shade 3 of the 5 columns.
Then shade 1 of the 2 rows.

What product of fractions
does the part shaded twice
show, and what is it?

Product: _____

Value: _____

True or false?

In a model of $\frac{3}{4} \times 8$, 8 is split into 4 equal parts and 3 of the parts are taken, so the product is 6.

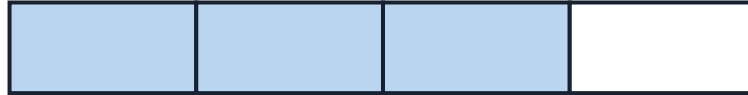
- A.** True
- B.** False

SET 66 • 5.NF.B.4.a

BACKWARD 1

Fractions of a whole in a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{2}{3}$ of a bar means 2 of 3 equal parts.

A. True

B. False

A bar is split into 6 equal parts and 4 parts are shaded.
What fraction is shaded?

A. $\frac{6}{4}$

B. $\frac{2}{6}$

C. $\frac{4}{6}$

D. $\frac{4}{10}$

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

True or false?

A bar split into 8 equal parts with 3 parts shaded shows $\frac{3}{5}$.

A. True

B. False

SET 66 • 5.NF.B.4.a

BACKWARD 2

Split a whole number into equal parts • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is split into 5 equal parts.

How much is in each part?

Answer: _____

True or false?

21 split into 3 equal parts gives 7 in each part.

A. True

B. False

Which expression shows 32 split into 4 equal parts?

A. $32 \div 4$

B. $32 - 4$

C. $4 \div 32$

D. 32×4

The tape diagram shows 16 split into 8 equal parts.
What number goes in each part?



Answer: _____

True or false?

36 split into 6 equal parts gives 5 in each part.

A. True

B. False

SET 66 • 5.NF.B.4.a

FORWARD 1

Use visual models to divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{6}$ s are in $\frac{2}{3}$?

Answer: _____

True or false?

$\frac{3}{4} \div \frac{1}{8} = 6$ because there are 6 eighths in $\frac{3}{4}$.

A. True

B. False

A model shows 2 wholes, each cut into thirds.

How many $\frac{1}{3}$ s are in 2?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. 5

D. 6

Divide.

$$\frac{5}{6} \div \frac{1}{12}$$

Quotient: _____

True or false?

$$\frac{2}{5} \div \frac{1}{10} = \frac{2}{50}$$

A. True

B. False

SET 66 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$\frac{3}{4} \times (-8)$$

Product: _____

True or false?

$$-\frac{2}{3} \times \frac{3}{4} = -\frac{1}{2}$$

A. True

B. False

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{3}{5}\right)$$

A. $-\frac{3}{10}$

B. $\frac{3}{10}$

C. $-\frac{4}{7}$

D. $\frac{4}{7}$

Find the product.

$$-\frac{2}{5} \times 15$$

Product: _____

True or false?

$$(-1) \times \left(-\frac{5}{6}\right) = -\frac{5}{6}$$

A. True

B. False

Find the area of a rectangle with fractional side lengths

MAIN

Grade 5 main question

5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares

3.MD.C.7.a

BACKWARD 2

The area formula for rectangles

4.MD.A.3

BACKWARD 4

Multiply a whole number by a fraction

4.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures

6.G.A.1

FORWARD 2

Area of composite figures

7.G.B.6

SET 32 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

Find the area of the rectangle.

Area: _____



A rectangle is $2\frac{1}{2}$ inches long and $\frac{4}{5}$ inch wide.

What is its area?

Area: _____

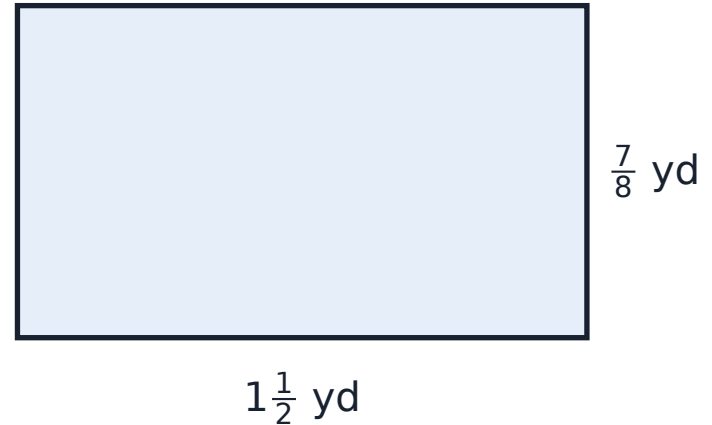
A rectangle that is $\frac{2}{3}$ unit by $\frac{3}{4}$ unit is tiled with small rectangles that are each $\frac{1}{3}$ unit by $\frac{1}{4}$ unit.

What is the area of the large rectangle?

- A.** $\frac{6}{12}$ square unit
- B.** $\frac{6}{7}$ square unit
- C.** $\frac{5}{7}$ square unit
- D.** $1\frac{5}{12}$ square units

Find the area of the rectangle.

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{2}$ foot has an area of 1 square foot.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares, with no gaps.

What is its area?

Area: _____

True or false?

Tiling a 3-by-5 rectangle takes 8 unit squares.

A. True

B. False

How many unit squares cover a 7-by-2 rectangle?

A. 72

B. 9

C. 14

D. 18

A rectangle has 5 rows of unit squares with 8 squares in each row.

What is its area?

Area: _____

True or false?

The number of unit squares that cover a rectangle equals its length times its width.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 2

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 5 m wide.

What is its area?

Area: _____

A rectangle has an area of 54 square feet and a length of 9 feet.

What is its width?

Width: _____

True or false?

The formula $A = l \times w$ gives the area of a rectangle.

A. True

B. False

A rectangle is 15 cm long and 4 cm wide.

What is its area?

- A.** 60 square cm
- B.** 45 square cm
- C.** 38 square cm
- D.** 19 square cm

Find the area of the rectangle.

Area: _____



SET 32 • 5.NF.B.4.b

BACKWARD 4

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

A rectangle is 3 units long and $\frac{2}{5}$ unit wide.

Write its area as a fraction.

Area: _____

True or false?

$$4 \times \frac{3}{8} = \frac{12}{8}$$

A. True

B. False

$$5 \times \frac{2}{3} = ?$$

A. $\frac{10}{3}$

B. $\frac{10}{15}$

C. $\frac{2}{15}$

D. $\frac{7}{3}$

Multiply.

$$6 \times \frac{3}{4}$$

Product: _____

True or false?

$$2 \times \frac{5}{6} = \frac{10}{12}$$

A. True

B. False

SET 32 • 5.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures • Grade 6 •
Geometry • 6.G.A.1

A triangle has a base of $4\frac{1}{2}$ inches and a height of 2 inches.

What is its area?

Area: _____

A parallelogram has a base of $3\frac{1}{4}$ feet and a height of 4 feet.

What is its area?

Area: _____

A triangle has a base of $\frac{3}{4}$ m and a height of $\frac{2}{3}$ m.

What is its area?

A. $\frac{1}{2}$ square meter

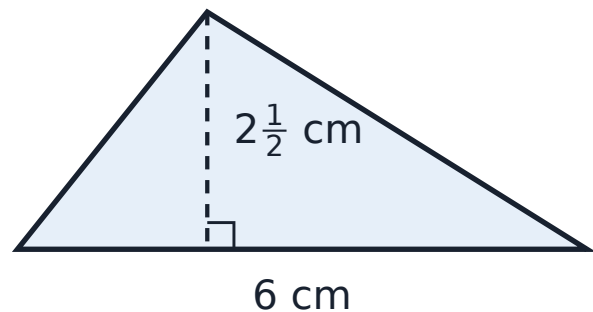
B. $\frac{5}{7}$ square meter

C. $\frac{1}{4}$ square meter

D. $\frac{6}{7}$ square meter

Find the area of the triangle.

Area: _____



True or false?

A trapezoid with bases of $2\frac{1}{2}$ m and $3\frac{1}{2}$ m and a height of 2 m has an area of 6 square meters.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

FORWARD 2

Area of composite figures • Grade 7 • Geometry • 7.G.B.6

A figure is made of a rectangle that is $6\frac{1}{2}$ ft by 4 ft and a triangle with a base of 4 ft and a height of 3 ft.

What is the total area?

Area: _____

A rectangular garden is 10 m by $7\frac{1}{2}$ m. A square patio with $2\frac{1}{2}$ -m sides is inside the garden.

What is the area of the garden that is not patio?

Area: _____

A $\frac{1}{2}$ -by-2 rectangle is cut out of a 5-by-4 rectangle.

What is the area of the part that is left?

- A.** 21 square units
- B.** 18 square units
- C.** 19 square units
- D.** 20 square units

A kite is made of two triangles. Each triangle has a base of 8 in. and a height of $3\frac{1}{2}$ in.

What is the area of the kite?

Area: _____

True or false?

A figure made of two $\frac{1}{2}$ -by-3 rectangles that do not overlap has an area of 6 square units.

- A. True
- B. False

Tile a rectangle with unit-fraction squares and connect the tiling to the area

MAIN	Grade 5 main question	5.NF.B.4.b
BACKWARD 1	Measure area by counting unit squares	3.MD.C.6
BACKWARD 2	Unit fractions of a length	3.NF.A.1
FORWARD 1	Pack a prism with unit-fraction cubes to find its volume	6.G.A.2
FORWARD 2	Area of composite figures with fractional measures	7.G.B.6

SET 67 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

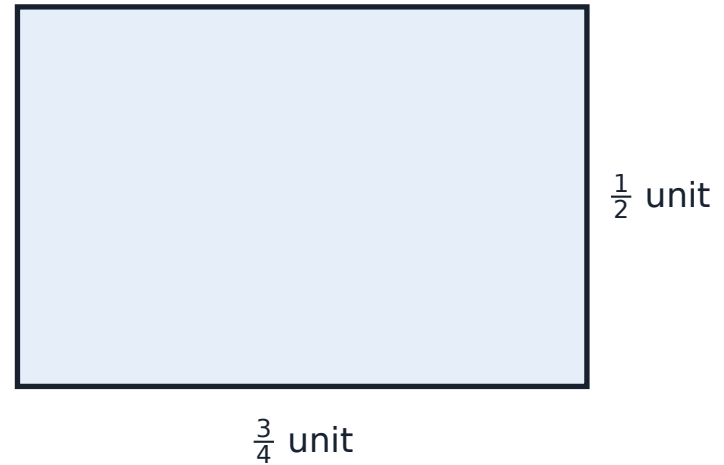
The rectangle is $\frac{3}{4}$ unit long and $\frac{1}{2}$ unit wide.

Draw lines to tile it with squares that are $\frac{1}{4}$ unit on each side.

How many squares fit, and what is the area?

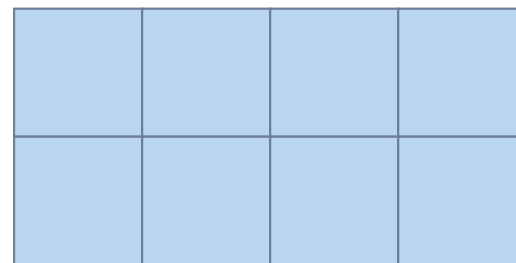
Squares: _____

Area: _____



The rectangle is tiled with squares that are $\frac{1}{3}$ unit on each side.

What is its area? Explain how the tiling shows it.



Area: _____

Explanation: _____

How many squares with $\frac{1}{6}$ -unit sides tile a rectangle that is $\frac{5}{6}$ unit by $\frac{1}{2}$ unit?

- A. 5
- B. 30
- C. 8
- D. 15

True or false?

Tiling a square with $\frac{2}{3}$ -unit sides takes 4 squares with $\frac{1}{3}$ -unit sides, so its area is $\frac{4}{9}$ square unit.

A. True

B. False

A rectangle is tiled with 3 rows of 4 squares. Each square has $\frac{1}{4}$ -unit sides.

What are the rectangle's side lengths and its area? Show that multiplying the side lengths gives the same area.

Side lengths: _____

Area: _____

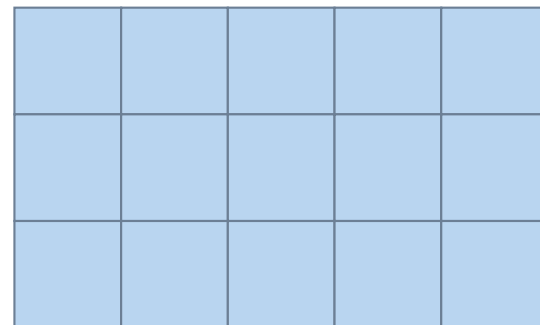
SET 67 • 5.NF.B.4.b

BACKWARD 1

Measure area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

Each small square is 1 square unit.

What is the area of the rectangle?



Area: _____

True or false?

A shape covered by 9 unit squares with no gaps or overlaps has an area of 9 square units.

- A.** True
- B.** False

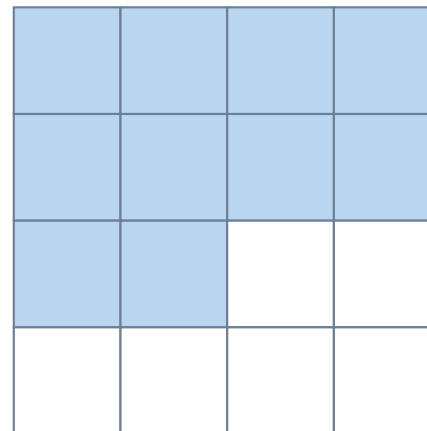
A rectangle is covered by 2 rows of 6 unit squares.

What is its area?

- A.** 8 square units
- B.** 6 square units
- C.** 12 square units
- D.** 16 square units

Each small square is 1 square inch.

What is the area of the shaded part?



Area: _____

True or false?

If the unit squares overlap, counting them still gives the exact area.

- A. True
- B. False

SET 67 • 5.NF.B.4.b

BACKWARD 2

Unit fractions of a length • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

A 1-unit side is split into 3 equal parts.

How long is each part?

Length: _____

True or false?

Four lengths of $\frac{1}{4}$ unit make 1 unit.

A. True

B. False

How many lengths of $\frac{1}{6}$ unit make $\frac{5}{6}$ unit?

A. 11

B. 5

C. 6

D. 1

A side is made of 3 lengths of $\frac{1}{4}$ unit.

How long is the side?

Length: _____

True or false?

Two lengths of $\frac{1}{8}$ unit make $\frac{2}{16}$ unit.

A. True

B. False

SET 67 • 5.NF.B.4.b

FORWARD 1

Pack a prism with unit-fraction cubes to find its volume • Grade 6 • Geometry •
6.G.A.2

A prism is 1 unit by $\frac{1}{2}$ unit by $\frac{1}{2}$ unit.

How many cubes with edges $\frac{1}{2}$ unit long pack it?

Cubes: _____

True or false?

A cube with edges $\frac{1}{2}$ unit long has a volume of $\frac{1}{8}$ cubic unit.

- A.** True
- B.** False

A box is packed with 24 cubes that each have edges $\frac{1}{2}$ unit long.

What is its volume?

- A.** 12 cubic units
- B.** 24 cubic units
- C.** 3 cubic units
- D.** 6 cubic units

Find the volume of a prism that is $\frac{3}{2}$ units by 1 unit by $\frac{1}{2}$ unit.

Volume: _____

True or false?

Packing a 1-by-1-by-1 cube with cubes that have edges $\frac{1}{3}$ unit long takes 9 cubes.

- A. True
- B. False

SET 67 • 5.NF.B.4.b

FORWARD 2

Area of composite figures with fractional measures • Grade 7 • Geometry •
7.G.B.6

A $\frac{1}{4}$ -m by $\frac{1}{3}$ -m rectangle is cut out of a $\frac{3}{4}$ -m by $\frac{2}{3}$ -m rectangle.

What area is left?

Area: _____

True or false?

Two rectangles that do not overlap, $\frac{1}{2}$ ft by 3 ft and $\frac{1}{2}$ ft by 1 ft, have a total area of 2 square feet.

A. True

B. False

A square with sides of $\frac{5}{2}$ units has a square with sides of $\frac{1}{2}$ unit removed.

What is the area of the figure?

A. 5 square units

B. 6 square units

C. 2 square units

D. $6\frac{1}{4}$ square units

A rug is $2\frac{1}{2}$ ft by 4 ft. A square corner with sides of $1\frac{1}{2}$ ft is cut off.

What is the area of the rug that is left?

Area: _____

True or false?

A triangle with a base of $\frac{3}{4}$ unit and a height of $\frac{2}{3}$ unit has an area of $\frac{1}{2}$ square unit.

A. True

B. False

Compare the size of a product to the size of one factor without multiplying

MAIN	Grade 5 main question	5.NF.B.5.a
BACKWARD 1	Multiplicative comparison	4.OA.A.2
BACKWARD 2	Comparing fractions to 1	4.NF.A.2
FORWARD 1	Scale quantities in ratio tables	6.RP.A.3.a
FORWARD 2	Use a scale to find lengths in scale drawings	7.G.A.1

SET 33 • 5.NF.B.5.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.a

Without multiplying, how does $\frac{3}{4} \times 20$ compare to 20?

- A.** It is less than 20.
- B.** It is equal to 20.
- C.** It is 4 times 20.
- D.** It is greater than 20.

True or false?

$5 \times 3\frac{1}{2}$ is greater than 5.

A. True

B. False

Without multiplying, is $\frac{2}{3} \times 15$ greater than, less than, or equal to 15?

Answer: _____

Which product is greater than 8?

A. $\frac{9}{8} \times 8$

B. $\frac{1}{2} \times 8$

C. $\frac{7}{8} \times 8$

D. $\frac{8}{8} \times 8$

A photo is 6 inches wide. It is resized to $\frac{5}{4}$ of its width.

Is the new width greater than or less than 6 inches?

Answer: _____

SET 33 • 5.NF.B.5.a

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.2

True or false?

3×12 is 3 more than 12.

A. True

B. False

A dog weighs 4 times as much as a 6-pound cat.
How much does the dog weigh?

Weight: _____

Which number is 5 times as much as 7?

A. 57

B. 12

C. 2

D. 35

True or false?

2×9 is less than 9.

A. True

B. False

Fill in the blank.

42 is ____ times as much as 7.

Blank: _____

SET 33 • 5.NF.B.5.a

BACKWARD 2

Comparing fractions to 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{4}{5}$ is greater than 1.

A. True

B. False

Which fraction is less than 1?

A. $\frac{6}{7}$

B. $\frac{9}{9}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

Write $>$, $<$, or $=$ to compare.

$$\frac{8}{8} \text{ ____ } 1$$

Symbol: _____

True or false?

$$\frac{3}{5} > 1$$

A. True

B. False

Is $\frac{11}{10}$ greater than, less than, or equal to 1?

Answer: _____

SET 33 • 5.NF.B.5.a

FORWARD 1

Scale quantities in ratio tables • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

A recipe uses 2 cups of rice for every 3 cups of water.
How many cups of water are needed for 8 cups of rice?

Cups of water: _____

A ratio table starts with 4 : 10.

Which ratio could also be in the table?

- A.** 4 : 5
- B.** 2 : 5
- C.** 8 : 15
- D.** 10 : 4

The table shows equivalent ratios.

Find the missing value.

Red	3	6	?
Blue	5	10	25

Missing value: _____

True or false?

Multiplying both numbers in $6 : 9$ by $\frac{1}{3}$ gives $2 : 3$, an equivalent ratio.

- A.** True
- B.** False

A map shows 5 km for every 2 cm.

How many kilometers do 7 cm show?

Kilometers: _____

SET 33 • 5.NF.B.5.a

FORWARD 2

Use a scale to find lengths in scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 inch = 4 feet. A wall is $3\frac{1}{2}$ inches long in the drawing.

How long is the real wall?

Length: _____

A room is 15 feet long.

How long is it on a drawing with a scale of 1 inch = 3 feet?

Length: _____

A drawing is reduced to $\frac{1}{2}$ of its size.

A 10-cm side becomes how long?

- A.** 2 cm
- B.** 20 cm
- C.** 10 cm
- D.** 5 cm

A drawing that is 4 cm by 6 cm is enlarged by a scale factor of $\frac{3}{2}$.

What are the new dimensions?

Dimensions: _____

True or false?

Using a scale factor less than 1 makes a drawing smaller.

A. True

B. False

Explain the effect of multiplying by a fraction greater than, less than, or equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 2	Fractions equal to 1	3.NF.A.3.c
BACKWARD 3	Multiply a whole number by a fraction	4.NF.B.4.b
FORWARD 1	Percents greater than and less than 100%	6.RP.A.3.c
FORWARD 2	Use a multiplier for percent increase and decrease	7.RP.A.3

SET 34 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

True or false?

Multiplying 12 by $\frac{7}{6}$ gives a product greater than 12.

- A.** True
- B.** False

Which statement explains why $\frac{4}{5} \times 30$ is less than 30?

- A.** $\frac{4}{5}$ is less than 1.
- B.** $\frac{4}{5}$ is greater than 1.
- C.** 30 is greater than 1.
- D.** The product is a fraction.

Explain why multiplying 24 by $\frac{5}{8}$ gives a product less than 24.

Explanation: _____

Which expression has a value equal to 9?

A. $\frac{6}{5} \times 9$

B. $\frac{5}{6} \times 9$

C. $\frac{1}{5} \times 9$

D. $\frac{5}{5} \times 9$

Without multiplying, order the products from least to greatest.

$$\frac{3}{4} \times 10, \frac{4}{3} \times 10, \frac{4}{4} \times 10$$

Order: _____

SET 34 • 5.NF.B.5.b

BACKWARD 2

Fractions equal to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

True or false?

$$\frac{4}{4} = 4$$

A. True

B. False

Write 1 as a fraction with a denominator of 6.

Fraction: _____

Which fraction is equal to 1?

A. $\frac{0}{8}$

B. $\frac{8}{1}$

C. $\frac{1}{8}$

D. $\frac{8}{8}$

True or false?

$\frac{6}{6}$ is greater than 1.

A. True

B. False

How many eighths make 1 whole?

Eighths: _____

SET 34 • 5.NF.B.5.b

BACKWARD 3

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

True or false?

$$3 \times \frac{2}{3} = 2$$

A. True

B. False

Multiply.

$$4 \times \frac{3}{4}$$

Product: _____

$$6 \times \frac{3}{2} = ?$$

A. 4

B. $\frac{9}{2}$

C. $\frac{18}{12}$

D. 9

Multiply.

$$5 \times \frac{2}{5}$$

Product: _____

True or false?

$$2 \times \frac{7}{8} = \frac{14}{16}$$

A. True

B. False

SET 34 • 5.NF.B.5.b

FORWARD 1

Percents greater than and less than 100% • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

True or false?

150% of 40 is less than 40.

A. True

B. False

What is 120% of 50?

Answer: _____

Which value is less than 80?

- A.** 200% of 80
- B.** 125% of 80
- C.** 75% of 80
- D.** 100% of 80

What is 100% of 37?

Answer: _____

What is 250% of 12?

Answer: _____

SET 34 • 5.NF.B.5.b

FORWARD 2

Use a multiplier for percent increase and decrease • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A \$60 price increases by 15%.

Which expression gives the new price?

A. 1.15×60

B. $60 + 15$

C. 0.85×60

D. 0.15×60

A \$90 jacket is 20% off.

Multiply by a decimal to find the sale price.

Sale price: _____

True or false?

Multiplying a price by 0.7 decreases it by 30%.

- A. True
- B. False

A population of 2,000 grows by 5%.

What is the new population?

Population: _____

Which multiplier shows a 40% decrease?

A. 0.6

B. 0.4

C. 1.4

D. 1.04

Explain equivalent fractions as multiplying by a fraction equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Equivalent fractions with a model	3.NF.A.3.b
FORWARD 1	Find equivalent ratios	6.RP.A.3.a
FORWARD 2	Test for a proportional relationship with equivalent ratios	7.RP.A.2.a

SET 68 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

Use the fact that $\frac{4}{4} = 1$ to explain why $\frac{2}{3} = \frac{8}{12}$.

Explanation: _____

Which fraction equal to 1 turns $\frac{3}{5}$ into $\frac{9}{15}$?

A. $\frac{3}{3}$

B. $\frac{15}{15}$

C. $\frac{9}{9}$

D. $\frac{5}{5}$

True or false?

$$\frac{5}{6} \times \frac{2}{2} = \frac{10}{12}, \text{ and } \frac{10}{12} = \frac{5}{6} \text{ because } \frac{2}{2} = 1.$$

A. True

B. False

What fraction equal to 1 can you multiply $\frac{7}{10}$ by to get $\frac{28}{40}$?

Fraction: _____

True or false?

$\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$ shows that $\frac{3}{4} = \frac{6}{12}$.

A. True

B. False

SET 68 • 5.NF.B.5.b

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

To write a fraction equal to $\frac{3}{5}$, you can multiply its numerator and denominator by the same nonzero whole number.

A. True

B. False

Find the missing number.

$$\frac{3}{5} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{7}{8}$?

A. $\frac{7}{16}$

B. $\frac{14}{24}$

C. $\frac{21}{24}$

D. $\frac{8}{7}$

Find the missing number.

$$\frac{1}{3} = \frac{4}{?}$$

Missing number: _____

True or false?

$$\frac{2}{9} = \frac{4}{27}$$

A. True

B. False

SET 68 • 5.NF.B.5.b

BACKWARD 2

Equivalent fractions with a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

True or false?

$$\frac{1}{2} = \frac{2}{4}$$

A. True

B. False

Find the missing number.

$$\frac{2}{3} = \frac{?}{6}$$

Missing number: _____

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{4}{3}$

D. $\frac{4}{6}$

The bar shows $\frac{1}{3}$ shaded. Each part is cut in half.

How many sixths are shaded?



Sixths: _____

True or false?

$$\frac{2}{4} = \frac{2}{8}$$

A. True

B. False

SET 68 • 5.NF.B.5.b

FORWARD 1

Find equivalent ratios • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Write a ratio equivalent to 3 : 5 by multiplying both numbers by 4.

Ratio: _____

True or false?

6 : 8 and 9 : 12 are equivalent ratios.

- A. True
- B. False

Which ratio is equivalent to 2 : 7?

- A.** 4 : 9
- B.** 7 : 2
- C.** 2 : 14
- D.** 6 : 21

Find the missing value.

$$5 : 6 = 20 : ?$$

Missing value: _____

True or false?

4 : 10 and 6 : 12 are equivalent ratios.

A. True

B. False

SET 68 • 5.NF.B.5.b

FORWARD 2

Test for a proportional relationship with equivalent ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	2	4	6
y	5	10	15

A. True

B. False

Which table shows a proportional relationship?

- A.** x: 1, 2, 3 and y: 4, 6, 8
- B.** x: 1, 2, 3 and y: 4, 8, 12
- C.** x: 1, 2, 3 and y: 3, 5, 7
- D.** x: 1, 2, 3 and y: 5, 6, 7

Are 3 : 4 and 12 : 16 equivalent ratios? Explain using a fraction equal to 1.

Answer: _____

Explanation: _____

True or false?

3 : 5 and 9 : 20 are equivalent ratios.

A. True

B. False

The ratios $2 : 9$ and $8 : y$ are equivalent.

Find y .

$y =$ _____

Solve real-world problems by multiplying fractions and mixed numbers

MAIN	Grade 5 main question	5.NF.B.6
BACKWARD 1	Word problems: a whole number times a fraction	4.NF.B.4.c
BACKWARD 2	Mixed numbers as fractions	4.NF.B.3.b
BACKWARD 4	Word problems that add fractions with like denominators	4.NF.B.3.d
FORWARD 1	Word problems with division of fractions	6.NS.A.1
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 35 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is 6 m long. $\frac{2}{3}$ of its length is planted with beans.

How many meters are planted with beans?

Meters: _____

Sam walks $\frac{5}{8}$ mile each day.

How far does he walk in 4 days?

A. $\frac{5}{32}$ mile

B. $\frac{9}{8}$ miles

C. $2\frac{1}{2}$ miles

D. $4\frac{5}{8}$ miles

A board is $3\frac{1}{3}$ feet long. Jo uses $\frac{3}{5}$ of the board.

How many feet does she use?

Feet: _____

A pitcher holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the pitcher?

Liters: _____

SET 35 • 5.NF.B.6

BACKWARD 1

Word problems: a whole number times a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each serving is $\frac{2}{3}$ cup.

How many cups are in 5 servings?

Cups: _____

True or false?

4 children each drink $\frac{3}{8}$ liter of milk. They drink $1\frac{1}{2}$ liters in all.

- A. True
- B. False

Each piece of ribbon is $\frac{5}{6}$ yard long.

How long are 3 pieces in all?

A. $\frac{5}{18}$ yard

B. $2\frac{1}{2}$ yards

C. $3\frac{5}{6}$ yards

D. $\frac{15}{18}$ yard

A bug crawls $\frac{3}{10}$ meter each minute.

How far does it crawl in 7 minutes?

Distance: _____

Each bag holds $\frac{1}{4}$ pound of nuts.

How many pounds are in 6 bags?

Pounds: _____

SET 35 • 5.NF.B.6

BACKWARD 2

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $2\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{1}{2} = \frac{11}{2}$$

A. True

B. False

Which fraction is equal to $4\frac{2}{5}$?

A. $\frac{22}{5}$

B. $\frac{42}{5}$

C. $\frac{11}{5}$

D. $\frac{24}{5}$

Write $\frac{15}{8}$ as a mixed number.

Mixed number: _____

SET 35 • 5.NF.B.6

BACKWARD 4

Word problems that add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

A recipe uses $\frac{3}{4}$ cup of oats. Ann makes the recipe twice.

How many cups of oats does she use? Write a fraction.

Cups: _____

True or false?

$$\frac{2}{5} + \frac{2}{5} + \frac{2}{5} = \frac{6}{5}$$

A. True

B. False

Ty walks $\frac{3}{8}$ mile to school and $\frac{3}{8}$ mile home.

How far does he walk in all?

A. $\frac{9}{8}$ mile

B. $\frac{6}{8}$ mile

C. $\frac{3}{8}$ mile

D. $\frac{6}{16}$ mile

A plant grew $\frac{1}{6}$ inch on Monday and $\frac{4}{6}$ inch on Tuesday.

How much did it grow in all?

Growth: _____

True or false?

$$\frac{5}{12} + \frac{5}{12} = \frac{10}{24}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 1

Word problems with division of fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{3}{4}$ -cup servings are in 6 cups of soup?

Servings: _____

A $\frac{7}{8}$ -mile trail is split into $\frac{1}{8}$ -mile sections.

How many sections are there?

Sections: _____

A rectangle has an area of $\frac{3}{4}$ square foot and a width of $\frac{1}{2}$ foot.

What is its length?

A. $\frac{3}{8}$ foot

B. $1\frac{1}{4}$ feet

C. $1\frac{1}{2}$ feet

D. $\frac{2}{3}$ foot

Ann has $2\frac{1}{2}$ pounds of clay. Each pot uses $\frac{5}{8}$ pound.

How many pots can she make?

Pots: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = \frac{1}{9}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot per second for $6\frac{1}{2}$ seconds.

What is the diver's change in depth?

Change: _____

The temperature was 2°F . It dropped $\frac{1}{2}^{\circ}\text{F}$ each hour for 9 hours.

What was the final temperature?

Temperature: _____

A stock loses $\frac{3}{8}$ dollar each day for 4 days.

What is the total change in value?

A. $-4\frac{3}{8}$ dollars

B. $-\frac{3}{32}$ dollar

C. $-1\frac{1}{2}$ dollars

D. $1\frac{1}{2}$ dollars

A tank contains $12\frac{1}{2}$ gallons of water. $\frac{2}{5}$ of the water drains out.

How many gallons drain out?

Gallons: _____

True or false?

$$-\frac{2}{3} \times 4\frac{1}{2} = -3$$

A. True

B. False

Solve real-world problems by multiplying mixed numbers

MAIN

Grade 5 main question

5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1

4.NF.B.3.b

BACKWARD 3

Breaking apart a factor to multiply

3.OA.B.5

BACKWARD 4

Measurement word problems with fractions

4.MD.A.2

FORWARD 1

Volume of prisms with fractional edge lengths

6.G.A.2

FORWARD 2

Surface area of prisms with fractional edge lengths

7.G.B.6

SET 36 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A bag of flour weighs $2\frac{1}{2}$ pounds.

How much do 3 bags weigh?

Weight: _____

A garden bed is $1\frac{1}{3}$ yards long and $1\frac{1}{2}$ yards wide.

What is its area?

Area: _____

A hiker walks $2\frac{1}{4}$ miles each hour.

How far does she walk in $1\frac{2}{3}$ hours?

A. $3\frac{1}{6}$ miles

B. $4\frac{1}{2}$ miles

C. $2\frac{2}{12}$ miles

D. $3\frac{3}{4}$ miles

A poster is $3\frac{1}{5}$ feet long. A copy is made $2\frac{1}{2}$ times as long.

How long is the copy?

Length: _____

True or false?

A recipe uses $1\frac{1}{2}$ cups of milk. Making $1\frac{1}{2}$ times the recipe uses $1\frac{1}{4}$ cups of milk.

A. True

B. False

SET 36 • 5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $1\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{13}{3}$ as a mixed number.

Mixed number: _____

True or false?

$$2\frac{2}{3} = \frac{22}{3}$$

A. True

B. False

Which mixed number is equal to $\frac{9}{2}$?

A. $2\frac{1}{9}$

B. $4\frac{1}{9}$

C. $9\frac{1}{2}$

D. $4\frac{1}{2}$

Write $5\frac{1}{2}$ as a fraction.

Fraction: _____

SET 36 • 5.NF.B.6

BACKWARD 3

Breaking apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking
• 3.OA.B.5

True or false?

$$4 \times (5 + 2) = 4 \times 5 + 4 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 12 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 3×15 ?

A. $3 \times 10 + 5$

B. $3 + 10 \times 5$

C. $3 \times 10 \times 5$

D. $3 \times 10 + 3 \times 5$

True or false?

$$7 \times 13 = 7 \times 10 + 3$$

A. True

B. False

Fill in the blank.

$$8 \times (4 + 1) = 8 \times 4 + 8 \times \underline{\quad}$$

Blank: _____

SET 36 • 5.NF.B.6

BACKWARD 4

Measurement word problems with fractions • Grade 4 • Measurement & Data •
4.MD.A.2

A ribbon is $\frac{3}{4}$ yard long.

How long are 4 ribbons in all?

Length: _____

True or false?

Each lap is $\frac{1}{2}$ mile. 6 laps is 3 miles.

A. True

B. False

A jug holds 2 liters.

How many milliliters are in 3 jugs?

- A.** 6,000 mL
- B.** 5,000 mL
- C.** 2,003 mL
- D.** 600 mL

Each box weighs $1\frac{1}{4}$ pounds.

How much do 2 boxes weigh?

Weight: _____

True or false?

3 pieces of wood, each $\frac{2}{5}$ meter long, make 1 meter in all.

- A. True
- B. False

SET 36 • 5.NF.B.6

FORWARD 1

Volume of prisms with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $2\frac{1}{2}$ ft by 2 ft by $1\frac{1}{2}$ ft.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its volume?

Volume: _____

A prism has a base area of $3\frac{1}{3}$ square cm and a height of $1\frac{1}{2}$ cm.

What is its volume?

- A.** $3\frac{1}{2}$ cubic cm
- B.** 10 cubic cm
- C.** $4\frac{5}{6}$ cubic cm
- D.** 5 cubic cm

Find the volume of the prism.

Volume: _____

